DISHA DUTTA

AI / Machine Learning Engineer · Open to relocation across India · Open to remote

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SUMMARY

Information Technology graduate who builds agentic LLM systems end to end — multi-agent workflows in LangGraph, hybrid retrieval pipelines, and FastAPI services with real databases, event queues and Docker behind them. Recent builds include a voice-first multi-agent shopping assistant and an explainable fraud and credit-risk engine for BFSI.

TECHNICAL SKILLS

Languages & CS: Python, SQL, C++, TypeScript, JavaScript; Data Structures & Algorithms, OOP, DBMS, Operating Systems
AI & Agents: LangGraph, LangChain, multi-agent orchestration, RAG, hybrid retrieval (BM25 + vector), human-in-the-loop workflows, prompt engineering, explainable AI, OpenAI API, Gemini API, NLP
ML: scikit-learn, TensorFlow, OpenCV, CNNs, classification, anomaly detection, feature engineering, model evaluation
Backend & Frontend: FastAPI, REST APIs, WebSockets, SQLAlchemy, JWT authentication, Streamlit, Next.js, Tailwind CSS
Data & Infra: PostgreSQL, MySQL, Redis, Docker, Docker Compose, Git, GitHub

EXPERIENCE

Icehut Technologies *Jan 2026 – Jun 2026*

AI/ML Intern · Location

- Built a real-time distracted-driver detection pipeline in Python: OpenCV for frame capture and face/ROI extraction, feeding a TensorFlow CNN that classified driver activity across *N* classes.
- Owned the training loop — cleaning, augmentation, class balancing, hyperparameter tuning — reaching *XX*% validation accuracy on held-out drivers.
- Tracked per-class precision and recall rather than overall accuracy, surfacing the classes the model confused; also prototyped LLM-assisted internal tooling (prompt design, structured output parsing).

Defence Research and Development Organisation (DRDO) *May 2025 – Jul 2025*

Web Development Intern · Lab / city

- Built a responsive internal web application (HTML, CSS, JavaScript, PHP, Bootstrap) on a MySQL/XAMPP stack for *what the app did*.
- Cut page load time by roughly 30% by compressing assets, deferring render-blocking scripts and removing redundant queries; verified across 5+ screen sizes and all browsers.

PROJECTS

Mitra AI — Voice-First Multi-Agent Shopping Assistant [code] [live] *2026*

Python · LangGraph · FastAPI · Next.js · PostgreSQL · Redis · Docker

- Voice-first grocery assistant for elderly users in India: accepts Hindi, Hinglish and English speech and turns loose spoken requests into structured purchase flows.
- Orchestrated eight LangGraph agents (intent, health guardrail, planning, search, comparison, decision, purchase, notification) with a human-in-the-loop pause before any purchase commits.
- Streamed per-agent progress to the UI over WebSockets with Redis event brokering; wrote orders behind an idempotency key plus a full PostgreSQL audit log.

Fraud Detection & Credit Risk AI [code] *Nov 2025*

Python · FastAPI · OpenAI API · Streamlit · MySQL · Docker

- Agentic BFSI system: a credit agent scores applications on traditional and alternative data (utility and rent history), returning a 0–1000 risk score routed to auto-approve, review or reject.
- A fraud agent scores transactions in real time on amount pattern, device fingerprint, location change and velocity abuse, pairing rule-based checks with LLM reasoning for cases rules miss.
- Every decision ships with its reasoning and contributing factors; an orchestrator agent sequences writes so a fraud case can never commit before its parent transaction.

Multi-Agent RAG System [code] *Dec 2025*

Python · LangGraph · LangChain

- Hybrid retrieval combining BM25 keyword search with vector similarity, so exact-term and semantic matches both surface for a query.
- A ranking agent re-ranks retrieved chunks before generation, cutting noisy context reaching the LLM; retrieval, ranking and generation are separate LangGraph nodes.

HireFlow AI — AI-Enabled HRMS [code] [live] *2026*

FastAPI · Next.js 14 · TypeScript · SQLAlchemy · Gemini API · Docker

- Full HR platform — employee directory, recruitment pipeline, leave and attendance, performance cycles, payroll and analytics — behind JWT auth with role-aware routes.
- Gemini-backed policy chatbot and candidate-comparison endpoints, with graceful fallback when no API key is set.

Also on GitHub: AI Resume Analyser (ATS scoring + AI rewrite, Streamlit) · AI Portfolio Builder · Scamify (phishing detector)

EDUCATION

B.Tech, Information Technology · Banasthali Vidyapith, Niwai, Rajasthan · *CGPA x.xx/10* *2022 – 2026*

Senior Secondary (Class XII) · LK Singhania Education Centre, Gotan, Rajasthan · *XX%* *2022*